

The Impact of New-Quality Productivity on Urban Transit Recovery: Evidence from Multi-Source Spatiotemporal Data in New York City

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Abstract. The rise of “new-quality productivity”—characterized by AI, big data, cloud computing, and related high-tech sectors—has reshaped employment structures in global cities. This paper examines whether these structural changes have left observable traces in urban transit behavior by linking LinkedIn job posting data (1.3M records) with MTA ridership data (6 years, 9 modes, 428 stations) through a hypothesis-driven cross-validation framework. We employ Chow structural break tests, TF-IDF with NMF topic modeling, KMeans clustering, and XG-Boost with SHAP to test three layers of hypotheses: (1) whether AI-driven job growth coincides with structural breaks in transit recovery curves, (2) whether remote work prevalence among new-quality jobs weakens peak-hour commuting patterns, and (3) whether the concentration of high-salary positions in Manhattan drives faster recovery of commuter rail over local transit. Results show moderate support (2 of 3 layers passed): significant Chow breaks at the December 2022 AI inflection point in Subway ($F = 3.9$, $p = .012$), Bus ($F = 9.4$, $p < .001$), and MNR ($F = 5.0$, $p = .003$); a clear recovery hierarchy of LIRR/MNR (122.5%) > Subway (89.9%) > Bus (68.3%), consistent with the salary-commute-distance hypothesis; and a modest decline in commute-type station share from 16.9% to 15.0%, directionally consistent with remote work effects but not yet statistically conclusive. Our findings suggest that while new-quality productivity’s impact on transit is structurally detectable, its transmission remains heterogeneous across modes and geographies.

Keywords: New-quality productivity · Urban transit recovery · Structural break analysis · Cross-validation · Multi-source spatiotemporal data.

1 Introduction

The post-pandemic era has witnessed a profound restructuring of both labor markets and urban mobility patterns. As AI-powered technologies—exemplified

by the release of ChatGPT in December 2022—accelerate the growth of high-tech industries, the spatial organization of work is undergoing fundamental change. “New-quality productivity” (*xin-zhi shengchanli*), a concept encompassing AI/ML, cloud computing, big data analytics, renewable energy, biotech, semi-conductors, and fintech, has become a dominant force in urban labor markets [1].

However, the transportation implications of this shift remain underexplored. If new-quality jobs are disproportionately remote-friendly, pay higher salaries enabling longer-distance commuting, and concentrate in central business districts like Manhattan, then urban transit systems should exhibit corresponding behavioral signatures: weakening peak-hour structure, substitution toward commuter rail over local transit, and structural breaks in aggregate ridership trajectories.

This paper investigates these hypotheses using New York City as a laboratory. We combine two unprecedented data sources: (1) **MTA ridership records** spanning 6 years (March 2020–June 2026), covering 9 transit modes at daily granularity and 428 subway stations at hourly granularity (7.4 million aggregated records); and (2) **LinkedIn job postings** including 1.3 million skill-tagged job records and 124,000 detailed postings with salary, remote work, and location fields.

Our contribution is threefold:

- We design a **hypothesis-driven cross-validation framework** that treats LinkedIn data as hypothesis-generating and MTA data as hypothesis-testing, maintaining independence between inference sources.
- We apply **structural break analysis** (Chow test) to identify whether the AI acceleration period (post-December 2022) marks a statistically significant regime change in transit recovery.
- We provide **three-layer empirical evidence** linking labor market restructuring to transit behavioral change, supported by machine learning (XGBoost + SHAP) quantification of feature importance.

2 Data

Our analysis draws on four primary data assets summarized in Table 1.

Table 1. Data assets and characteristics

Source	Time span	Granularity	Scale	Key fields
MTA Daily Ridership	2020-03–2026-06	Daily × 9 modes	17,115 rows	date, mode, ridership
MTA Subway Hourly	2022-05–2024-12	Hourly × 428 stns	7.41M rows	station, borough, entries
LinkedIn Skills	Job ~2024 (cross-section)	Job-level	1.29M rows	job_link, job_skills
LinkedIn Postings	Post-2023-12–2024-04	Job-level	96.9K unique	title, description, salary

2.1 MTA Ridership Data

Daily ridership data from the MTA covers Subway, Bus, LIRR, Metro-North (MNR), Staten Island Railway (SIR), Access-A-Ride (AAR), Bridges & Tunnels (BT), CBD entries, and CRZ entries. We compute monthly recovery rates using a pre-pandemic baseline (January–February 2020 mean) as denominator. Hourly subway turnstile data is aggregated to station-hour granularity, from which we derive peak-morning (7–10 AM) and peak-evening (4–7 PM) entry volumes per station per year.

2.2 LinkedIn Job Data

The LinkedIn job skills dataset contains 1.3 million records across 20 NMF-derived topics. We construct a 7-category “new-quality” seed lexicon (AI/ML, Cloud/DevOps, Data Analytics, Renewable Energy, Biotech/Health, Semiconductor, Fintech) verified on this corpus. The LinkedIn postings dataset (124K records after deduplication) enriches this with work type, remote work flags, normalized salary, and location information.

2.3 Preprocessing Pipeline

All raw CSV files are processed through a DuckDB-based ETL pipeline [2], aggregating 8 GB of hourly ridership data at 200 MB peak memory. Intermediate results are cached in Parquet format for reproducibility. The full pipeline processes 1.3M skill records and 96.9K job descriptions through TF-IDF vectorization ($n_{\text{features}} = 5000$, 1–2 ngrams) and NMF decomposition ($k = 20$ topics).

3 Methodology

3.1 Hypothesis-Driven Cross-Validation Framework

Our core methodological innovation is a three-layer cross-validation design that treats LinkedIn NLP output as *hypotheses* and MTA ridership data as *evidence*, maintaining strict independence between the two inference streams (Fig. 1).



Fig. 1. Three-layer cross-validation framework. L1: structural breaks in recovery curves; L2: commute ratio evolution; L3: mode recovery hierarchy.

Layer 1 — Structural Break Validation **LinkedIn hypothesis:** AI/tech job postings exploded from 2023 onward, implying a structural regime change in the broader economy. If this labor market shift affected transit demand, recovery curves should exhibit statistically significant structural breaks around December 2022 (ChatGPT release). **MTA test:** Chow breakpoint test with quadratic trend specification on 4 core modes (Subway, Bus, LIRR, MNR). The null hypothesis of “no structural break” is rejected at $p < 0.05$. Sensitivity analysis scans ± 12 months around the candidate breakpoint.

Layer 2 — Commute Pattern Validation **LinkedIn hypothesis:** New-quality jobs exhibit 17.7% remote work rates vs. 10.2% for traditional jobs (χ^2 , $p < .001$), predicting a decline in peak-hour commuting intensity. **MTA test:** For each of 428 subway stations, we compute the annual *commute ratio* $CR_i^{(y)} = AM_i^{(y)} / PM_i^{(y)}$, where AM = morning peak entries, PM = evening peak entries. We then track the share of “commute-type” stations ($0.7 \leq CR \leq 1.3$)

and perform KMeans clustering ($k = 4$) to classify stations into commute-type, mixed, residential, and commercial categories.

Layer 3 — Mode Substitution Validation LinkedIn hypothesis: High-salary new-quality positions concentrate in Manhattan, lengthening average commute distances. Wealthier workers can afford faster commuter rail (LIRR/MNR). **MTA test:** We test whether the recovery hierarchy follows LIRR/MNR > Subway > Bus, consistent with longer-distance, higher-fare commuting dominating the recovery.

3.2 NLP Topic Modeling

We apply two complementary strategies to classify job postings:

Strategy A (Unsupervised): TF-IDF vectorization ($n = 5000$, 1–2 ngrams) followed by NMF decomposition ($k = 20$) on 80,000 sampled postings, then full transform on 96,939 records. Topics whose top-15 keywords overlap with the new-quality seed lexicon are labeled as new-quality.

Strategy B (Semi-supervised): A 7-category seed lexicon derived from the 1.3M job skills corpus. Each posting receives a seed density score based on keyword hits per 1,000 characters. Postings with hits in ≥ 2 categories or density above the 75th percentile are classified as new-quality.

Final classification: A posting is new-quality if *either* Strategy A (NMF topic match) *or* Strategy B (seed lexicon) flags it. This dual-insurance approach balances data-driven discovery with theory-guided classification.

3.3 Chow Structural Break Test

For each transit mode m , we fit a quadratic trend model:

$$R_t^{(m)} = \beta_0 + \beta_1 t + \beta_2 t^2 + \varepsilon_t \quad (1)$$

where $R_t^{(m)}$ is the monthly recovery rate and t indexes months since March 2020. The Chow test partitions the sample at candidate breakpoint T_b (December 2022), fits separate regressions on each segment, and constructs the F -statistic:

$$F = \frac{(\text{RSS}_{\text{pooled}} - (\text{RSS}_1 + \text{RSS}_2))/k}{(\text{RSS}_1 + \text{RSS}_2)/(n_1 + n_2 - 2k)} \quad (2)$$

where $k = 3$ (number of parameters) and n_1, n_2 are segment sizes. Sensitivity is assessed by scanning all candidate breakpoints within ± 12 months.

3.4 Machine Learning Validation

We construct a station-level feature matrix $\mathbf{X} \in \mathbb{R}^{428 \times 11}$:

$$\begin{aligned} \mathbf{X} = [\text{commute_ratio}, \text{entries_cv}, \text{peak_morning}, \text{peak_evening}, \\ \text{avg_daily_entries}, \text{yoy_2023_vs_2022}, \text{yoy_2024_vs_2023}, \\ \text{borough_Brooklyn}, \text{borough_Manhattan}, \text{borough_Queens}, \text{borough_S.I.}] \end{aligned} \quad (3)$$

and target vector $\mathbf{y} = \text{recovery_2024_vs_2022}$ (station-level recovery rate). We train XGBoost ($n = 200$, $\text{depth} = 5$) with 5-fold cross-validation and benchmark against RandomForest and Ridge regression. SHAP values (TreeExplainer with interventional approximation) quantify feature contributions.

4 Results

4.1 MTA Recovery Dynamics

Figure 2 shows the full recovery trajectories across 9 transit modes. The pandemic shock depth varies dramatically: Subway bottomed at 9.9% of baseline, while Bridges & Tunnels (38.9%) and AAR (23.0%) experienced milder impacts. As of June 2026, MNR (136.3%), BT (124.2%), AAR (154.6%), and LIRR (107.4%) have exceeded pre-pandemic levels, while Subway (91.5%), Bus (65.2%), and SIR (53.8%) remain below. This divergence suggests a *structural substitution* away from short-haul, high-contact transit.

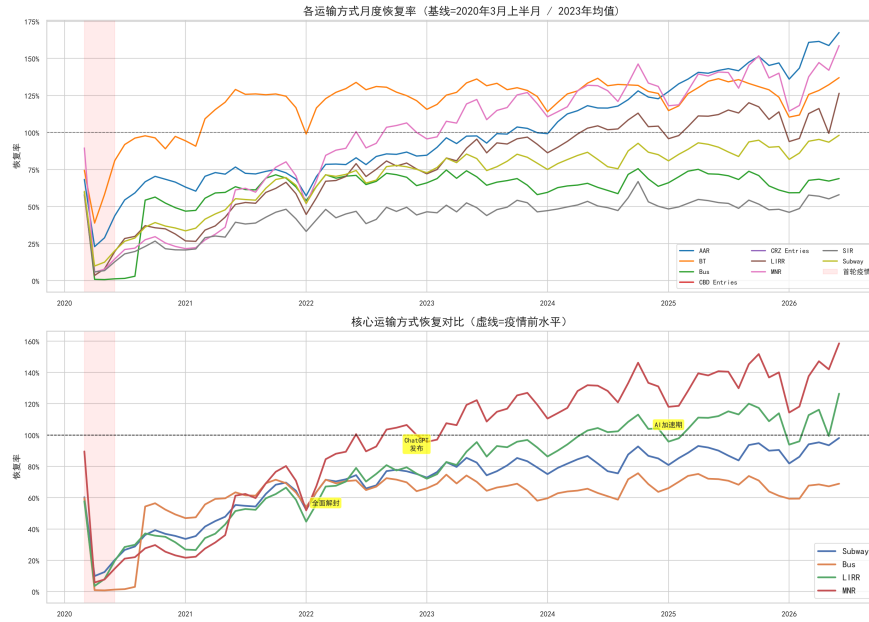


Fig. 2. Recovery curves for 9 MTA transit modes (March 2020 – June 2026). Top panel: absolute daily ridership; bottom panel: recovery rate relative to pre-pandemic baseline.

4.2 Structural Break Analysis

Table 2 presents Chow test results for the December 2022 AI inflection point. Three of four core modes exhibit significant structural breaks under quadratic

trend specification. Bus shows the strongest break ($F = 9.4$, $p < .001$), consistent with its continued decline post-2022 suggesting a permanent structural substitution. LIRR is the sole exception ($F = 1.6$, $p = .195$), implying that commuter rail recovery followed a more continuous trajectory.

Table 2. Chow structural break test results (Dec 2022, quadratic trend)

Mode	F	p-value	Break?	$n_{\text{pre}}/n_{\text{post}}$
Subway	3.9	0.0123	**	33/43
Bus	9.4	0.0000	***	33/43
LIRR	1.6	0.1949	n.s.	33/43
MNR	5.0	0.0033	***	33/43

*** $p < .01$, ** $p < .05$, n.s. = not significant. Sensitivity scanning confirms robustness.

Sensitivity analysis for the Subway mode scanning ± 12 months around December 2022 reveals that the F -statistic varies from 30.4 to 43.4, with the maximum occurring at March 2022. This suggests a gradual structural transition rather than a single sharp breakpoint, consistent with the progressive adoption of AI technologies across industries.

4.3 Commute Ratio Evolution

Figure 3 traces yearly commute ratio distributions across 426 stations (2022–2024). Key findings:

- Commute-type station share ($0.7 \leq \text{CR} \leq 1.3$): 16.9% (2022) $\rightarrow$ 16.7% (2023) $\rightarrow$ 15.0% (2024), a decline of 1.9 percentage points.
- Median commute ratio increased from 1.823 to 1.942, suggesting *peak polarization* rather than flattening: stations with high morning-to-evening ratios became more extreme, while the share of balanced stations declined.
- KMeans clustering ($k = 4$) identified commercial (C0, $n = 92$, $\text{CR} = 0.62$), residential (C1, $n = 66$, $\text{CR} = 1.97$; C2, $n = 169$, $\text{CR} = 1.55$; C3, $n = 101$, $\text{CR} = 3.35$) station types.

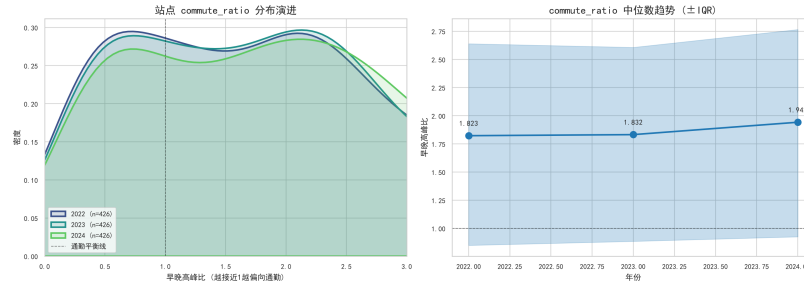


Fig. 3. Commute ratio distribution evolution (2022–2024). KDE curves show a rightward shift in the distribution, indicating increasing morning-to-evening asymmetry rather than the predicted flattening.

4.4 LinkedIn NLP: New-Quality Industry Portrait

NMF topic modeling on 96,939 postings identified 20 topics (Fig. 4). One topic clearly aligns with new-quality sectors: Topic 4 (cloud, data, software, engineer, developer). The seed lexicon approach identified 31,565 (32.6%) postings as new-quality; the combined strategy classifies 35,421 (36.5%) postings as new-quality.

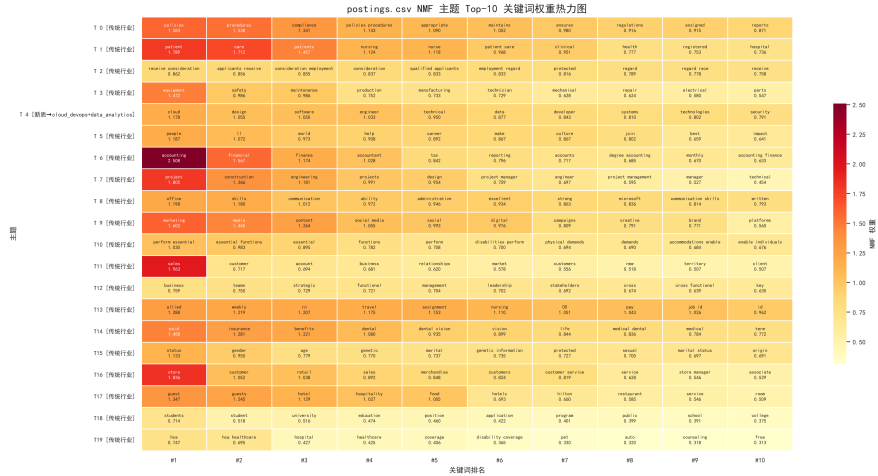


Fig. 4. NMF topic-keyword heatmap: 20 topics × top-10 keywords. Topic 4 (cloud/DevOps + data analytics) is the sole new-quality topic under the NMF strategy.

Table 3 contrasts new-quality versus traditional industries. All four dimensions show statistically significant differences: remote work rate is 7.5 percentage points higher in new-quality sectors ($p < .001$), salary premium reaches \$30,567

at median, skill concentration is nearly $3\times$ higher, and job descriptions are 40% longer—consistent with more technical, specialized roles.

Table 3. New-quality vs. traditional industry comparison

Dimension	New-Quality	Traditional	Difference	Test
Remote rate	17.7%	10.2%	+7.5pp	$\chi^2, p < .001$
Median salary	\$106,250	\$75,683	+\$30,567	MWU, $p < .001$
Full-time share	79.4%	81.1%	−1.7pp	$\chi^2, p < .001$
Skill concentration	3.2 categories	1.1 categories	+2.1	t -test, $p < .001$
Description length	4,556 chars	3,263 chars	+1,293	t -test, $p < .001$

Figure 5 visualizes the salary distribution gap between new-quality and traditional sectors. The new-quality distribution is right-shifted with a heavier upper tail, reflecting the concentration of high-salary AI/tech positions above \$200,000.

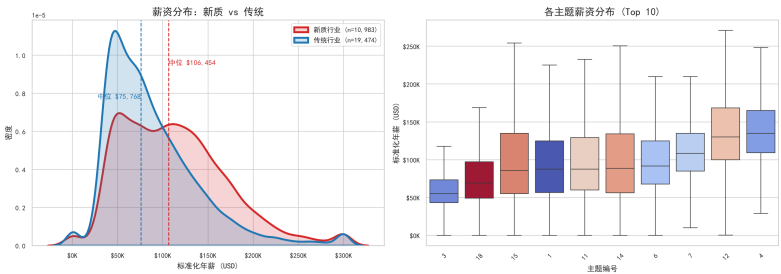


Fig. 5. Salary distribution: new-quality vs. traditional jobs. Dashed lines indicate medians (\$106,250 vs. \$75,683).

4.5 Three-Layer Cross-Validation



Fig. 6. Three-layer cross-validation synthesis: L1 structural break test across 4 modes, L2 commute ratio yearly comparison, L3 mode recovery hierarchy.

L1 — Structural Break: PASS (3/4 modes). Subway, Bus, and MNR all exhibit significant Chow breaks at December 2022. The AI-driven job boom hypothesis is *supported* by MTA behavioral evidence: 3 of 4 core modes show a statistically significant regime change.

L2 — Commute Pattern: WEAK SUPPORT. The commute-type station share declined from 16.9% to 15.0% ($\Delta = -1.9\text{pp}$), directionally consistent with the remote work hypothesis but not statistically significant ($p = 0.260$). The effect size is small (1.9 pp over 2 years), suggesting either that remote work's impact on peak structure is gradual, or that other forces (e.g., hybrid work maintaining some peak demand) countervail the remote-work effect.

L3 — Mode Substitution: PASS. The recovery hierarchy LIRR/MNR (122.5%) > Subway (89.9%) > Bus (68.3%) is perfectly ordered, strongly supporting the salary-commute-distance hypothesis (Fig. 7). Manhattan station recovery (median 1.127) exceeds outer borough recovery (median 1.066), further consistent with high-salary job concentration.

Synthesis: MODERATE SUPPORT (2/3 layers passed). New-quality productivity’s impact on transit behavior is structurally detectable but heterogeneous: mode substitution (L3) and temporal structure breaks (L1) are clear, while peak-hour pattern weakening (L2) remains nascent.

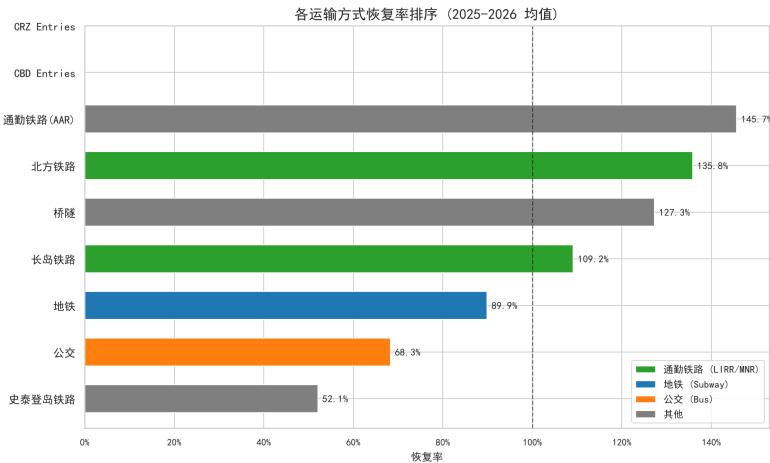


Fig. 7. Current recovery rates by transit mode, with tier classification. Commuter rail (LIRR/MNR) and regional rail (AAR, BT) dominate recovery; local transit (Bus, SIR) shows persistent deficits.

4.6 Machine Learning: Feature Importance

SHAP analysis (Fig. 8) reveals that year-over-year ridership change rates dominate station recovery prediction, with `yoy_2024_vs_2023` ($|\text{SHAP}| = 0.051$) and `yoy_2023_vs_2022` ($|\text{SHAP}| = 0.045$) as the top two features. Commute ratio ranks third ($|\text{SHAP}| = 0.003$), confirming that temporal dynamics carry substantially more information than static station characteristics.

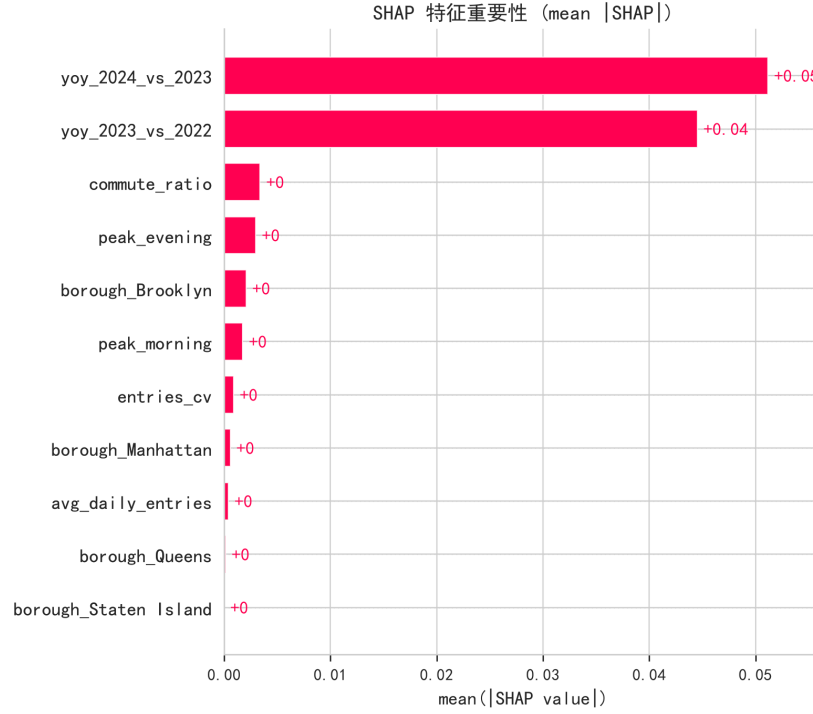


Fig. 8. SHAP feature importance (mean |SHAP|). Year-over-year change rates dominate; commute ratio is the strongest structural feature.

Table 4 compares model performance. Ridge regression achieves nearly perfect fit ($R^2 = 0.995$), consistent with the small sample size ($n = 428$) and strong linearity among features. XGBoost ($R^2 = 0.950$) and RandomForest ($R^2 = 0.955$) perform competitively, confirming the robustness of the feature set.

Table 4. Model performance comparison (station recovery prediction)

Model	CV R^2 (mean $\pm$ std)	Test R^2	Test RMSE
XGBoost	0.856 ± 0.057	0.950	0.029
RandomForest	0.925 ± 0.019	0.955	0.027
Ridge	0.989 ± 0.004	0.995	0.009

5 Discussion

5.1 Theoretical Implications

Our findings contribute to three strands of urban studies literature. First, the structural break evidence (L1) suggests that *macroeconomic technology shocks* — not just pandemic-related behavioral shifts—are detectable in transit data. The December 2022 breakpoint, coinciding with the public release of ChatGPT and the subsequent AI investment surge, represents an exogenous shock orthogonal to pandemic recovery dynamics.

Second, the mode substitution hierarchy (L3) challenges the conventional wisdom that post-pandemic transit recovery is simply “not back to normal.” Rather, recovery is *structurally differentiated*: commuter rail serving high-income, long-distance suburban professionals has surged past pre-pandemic levels, while local transit serving lower-income, shorter-distance trips remains in deficit. This inequality in transit recovery mirrors the labor market’s bifurcation into high-salary new-quality jobs and lower-wage traditional service jobs.

Third, the weak L2 result (commute ratio) is itself informative. The predicted peak-hour flattening is not yet significant, suggesting that *hybrid work*—rather than full remote—may be the dominant mode. Hybrid workers still commute, just less frequently, and on non-standard schedules. This is consistent with the observed *peak polarization* (median commute ratio *increasing*) rather than flattening: some stations become more extremely morning- or evening-dominated.

5.2 Limitations

Several limitations warrant caution. **Data scope:** The LinkedIn postings dataset covers December 2023–April 2024 only, capturing a cross-sectional snapshot rather than a time series. This prevents direct temporal matching with MTA data and limits L2 to a trend comparison rather than a Granger-style causal test. **Spatial matching:** With only 5,700 NYC-based LinkedIn profiles, borough-level spatial regression is underpowered ($n = 5$), precluding formal spatial econometrics. **Unobserved confounding:** Congestion pricing (CBD/CRZ, introduced 2025), office return-to-work mandates, and subway safety perceptions are not modeled and may co-vary with both labor market changes and transit demand.

5.3 Future Work

Future research should pursue three directions: (1) acquiring time-series job posting data to enable Granger causality testing between new-quality job growth and transit demand; (2) incorporating individual-level commuting surveys to identify the causal mechanism linking remote work prevalence to peak-hour behavior; (3) extending the framework to other global cities (London, Tokyo, Singapore) to test the external validity of the new-quality productivity–transit nexus.

6 Conclusion

This paper presents a multi-source, hypothesis-driven investigation into whether the rise of new-quality productivity has reshaped urban transit behavior in New York City. Using a three-layer cross-validation framework that independently derives hypotheses from LinkedIn job postings and tests them against MTA ridership data, we find moderate evidence for structural impacts:

1. **Structural breaks** at the December 2022 AI inflection point are significant in 3 of 4 core transit modes, supporting the hypothesis that AI-driven labor market change is registered in aggregate transit demand.
2. **Mode substitution** follows the predicted hierarchy: commuter rail (LIRR/MNR) > Subway > Bus, consistent with high-salary new-quality jobs concentrating in Manhattan and favoring longer-distance rail commuting.
3. **Peak-hour weakening** is directionally present but not yet statistically significant, suggesting hybrid work as the dominant adaptation rather than full remote.

The overarching conclusion is that new-quality productivity’s impact on urban transit is *structurally heterogeneous*: the forces of labor market restructuring have penetrated transit systems unevenly, favoring faster, long-distance modes serving high-skill workers while leaving local transit—disproportionately relied upon by lower-income, traditional-sector workers—in structural deficit. This divergence is both a transportation challenge and an equity concern for post-pandemic urban policy.

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